11.5 Monitoring a cache

CloudWatch is the service on AWS that stores all kinds of metrics. Like other services, ElastiCache nodes send metrics to CloudWatch as well. The most important metrics to watch follow:

- `CPUUtilization`—The percentage of CPU utilization.
- `SwapUsage`—The amount of swap used on the host, in bytes. *Swap* is space on disk that is used if the system runs out of physical memory.
- `Evictions`—The number of nonexpired items the cache evicted due to the memory limit.
- `ReplicationLag`—This metric is applicable only for a Redis node running as a read replica. It represents how far behind, in seconds, the replica is in applying changes from the primary node. Usually this number is very low.

In this section we'll examine those metrics in more detail and give you some hints about useful thresholds for defining alarms on those metrics to set up production-ready monitoring for your cache.

11.5.1 Monitoring host-level metrics

The virtual machines running underneath the ElastiCache service report CPU utilization and swap usage. CPU utilization usually gets problematic when crossing 80–90%, because the wait time explodes. But things are more tricky here. Redis is single-threaded. If you have many cores, the overall CPU utilization can be low, but one core can be at 100% utilization. Use the `EngineCPUUtilization` metric to monitor the Redis process CPU utilization. Swap usage is a different topic. You run an in-memory cache, so if the virtual machine starts to swap (move memory to disk), the performance will suffer. By default, ElastiCache for Memcached and Redis is configured to limit memory consumption to a value smaller than what's physical available (you can tune this) to have room for other resources (e.g., the kernel needs memory for each open socket). But other processes (such as kernel processes) are also running, and they may start to consume more memory than what's available. You can solve this issue by increasing the memory of the cache, either by increasing the node type or by adding more shards.